

Marcin Copik

SERVERLESS · HPC · PHD RESEARCHER

✉ mcopik@gmail.com | 🏠 <https://mcopik.github.io/> | 📄 <https://github.com/mcopik>

Summary

I work at the intersection of cloud computing and HPC. My PhD focused on serverless programming models designed to bridge supercomputers and data centers. I developed tailored solutions for different levels of the FaaS computing stack: from hardware and network optimizations to efficient system designs and performance modeling.

Education

PhD in Computer Science

ETH ZÜRICH

April 2018 - March 2024

Zürich, Switzerland

- Thesis: High-Performance Serverless for HPC and Clouds
- Advisor: Prof. Torsten Hoefler

Master of Science (MSc) in Simulation Sciences

RWTH AACHEN

September 2014 - July 2017

Aachen, Germany

- Grade: 1.5. Interdisciplinary program. Major subject: High-Performance Computing
- Thesis: Parallel Prefix Algorithms for the Registration of Arbitrarily Long Electron Micrograph Series
- Advisor: Prof. Paolo Bientinesi, Prof. Benjamin Berkels

Summer School in Mathematics

UNIVERSITY OF PERUGIA

August 2014

Perugia, Italy

- Courses: Stochastic Processes, Functional Analysis

Bachelor of Science in Engineering (BSc) in Computer Science

SILESIA UNIVERSITY OF TECHNOLOGY

September 2010 - March 2014

Gliwice, Poland

- Grade 5(A). Major subject: Software Engineering
- Thesis: GPU-accelerated stochastic simulator engine for PRISM model checker
- Advisor: Prof. Tadeusz Czachorski
- Additionally, finished two of three years of BSc program in Mathematics (2012-2014).

Experience

Postdoctoral Researcher

ETH ZÜRICH

Zürich, Switzerland

May 2024 -

- Advising for Bachelor and Master thesis projects.
- Conducting interviews for PhD and PostDoc candidates.
- Teaching assistant for Bachelor and Master courses.

Research Assistant

ETH ZÜRICH

Zürich, Switzerland

April 2018 - April 2024

- Advising for Bachelor and Master thesis projects.
- Conducting interviews for PhD and PostDoc candidates.
- Teaching assistant for Bachelor and Master courses.

Research Intern

MICROSOFT

Redmond, WA, USA

June - October 2019

- Analyzing microarchitectural implications of serverless workloads.
- Supervisor: Bobbie Manne.

Mentor

GOOGLE SUMMER OF CODE

STELLAR Group, SPCL.

2017, 2018, 2023, 2024

- Students: Ajai V George, Gabriel Laberge (co-mentored), Matt Nappo, Abhishek Kumar, Syed Mujtaba, Prajin Khadka.

Student Research Assistant

RWTH AACHEN, HIGH-PERFORMANCE AND AUTOMATIC COMPUTING

Aachen, Germany

2016 - December 2017

- Benchmarking linear algebra frameworks.
- Supervisor: Prof. Paolo Bientinesi.

Research Assistant

LOUISIANA STATE UNIVERSITY, STE||AR GROUP

- Integrating single-source GPU programming in HPX.
- Supervisor: Prof. Hartmut Kaiser.

Baton Rouge, LA, USA

April 2016 - August 2016

Student Research Assistant

JÜLICH SUPERCOMPUTING CENTRE

- Developing tools for performance analysis of parallel applications at Scalasca.
- Supervisor: Dr Pavel Saviankou.

Jülich, Germany

October 2014 - March 2016

Software Engineer

GOOGLE SUMMER OF CODE

- Integrating single-source GPU programming in HPX.
- Supervisor: Dr Hartmut Kaiser.

Organization: The STE||AR Group

2015

Software Engineer

GOOGLE SUMMER OF CODE

- Improving statistical model checking.
- Supervisor: Dr Vojtěch Forejt, Dr Dave Parker.

Organization: PRISM model checker

2014

Student Research Assistant

THE INSTITUTE OF THEORETICAL AND APPLIED INFORMATICS

- Implementing GPU simulator of Markov Chains.
- Supervisors: Dr Mateusz Nowak, Dr Artur Rataj.

Gliwice, Poland

2012 - 2013

Student Research Assistant

SILESIA UNIVERSITY OF TECHNOLOGY

- Implementing algorithms for registration of respiratory motion.
- Supervisor: Dr Dominik Spinczyk.

Gliwice, Poland

2012 - 2013

Honors & Awards

2025	SIGHPC Doctoral Dissertation Award , ACM SIGHPC
2025	Best Paper Award Nominee , ACM/IEEE Supercomputing 2025
2025	Gordon Bell Award in Climate Modeling , ACM/IEEE Supercomputing 2025
2024	Best Research Poster Award , ACM/IEEE Supercomputing 2024
2023, 2024	SIGHPC Travel Grant , awarded for travel to ACM/IEEE Supercomputing 2023 & 2024
2022	ACM/IEEE George Michael Memorial HPC Fellowship , awarded for contributions into high-performance serverless
2022	Gold Medal at the ACM Student Research Competition , ACM/IEEE Supercomputing 2022
2022	AWS Cloud Credit for Research Application
2022	Google Cloud Research Credits
2021	Microsoft Research PhD Fellowship , awarded for the 2021/2022 academic year
2019	Gold Medal at the ACM Student Research Competition , ACM/IEEE Supercomputing 2019

Major Projects

SeBS: Serverless Benchmark Suite

FAAS BENCHMARKING SUITE WITH AUTOMATIC BUILD, DEPLOYMENT, AND MEASUREMENTS. GITHUB: 175 STARS, 85 FORKS.

GOOGLE SCHOLAR: 231 CITATIONS.

Python, Cloud, FaaS

XaaS: Performance-Portable HPC Containers

RECONFIGURABLE CONTAINERS FOR PERFORMANCE PORTABILITY. BEST PAPER AWARD FINALIST AT SUPERCOMPUTING 2025.

Containers, LLVM, C++

rFaaS: RDMA-Enabled FaaS Platform

HIGH-PERFORMANCE SERVERLESS PLATFORM WITH RDMA ACCELERATION FOR HPC FUNCTION INVOCATIONS. GITHUB: 53

STARS, 18 FORKS. GOOGLE SCHOLAR: 45 CITATIONS.

C++, RDMA, FaaS

Peer-reviewed Publications

Extra-P – Empirical Performance Modeling Made Easy

CALOTOIU A., **COPIK M.**, CZAPPA F., GEISS A., MORAIS G., RITTER M., SHUDLER S., HOEFLER T., WOLF F.

Frontiers in High-Performance
Computing

2026

SecPerf: Demystifying Cost of Confidential HPC

CHRAPEK M., S. SHEN, P. IFF, T. CHEN, M. KHALILOV, **COPIK M.**, BESTA M., HOEFLE T.

IPDPS

2026

XaaS Containers: Performance-Portable Representation With Source and IR Containers

COPIK M., ALNUAIMI E., KAMATAR A., HAYOT-SASSON V., MADONNA A., GAMBLIN T., CHARD K., FOSTER I., HOEFLE T.

- Acceptance Rate 22% (137/623), **Best Paper Award Finalist (6/137)**

Supercomputing

2025

Computing the Full Earth System at 1km Resolution

KLOCKE D.. [AND 25 OTHERS, INCLUDING **COPIK M.**]

- **Gordon Bell Award in Climate Modeling**

Supercomputing

2025

Core Hours and Carbon Credits: Incentivizing Sustainability in HPC

KAMATAR A., GONTHIER M., HAYOT-SASSON V., BAUER A., **COPIK M.**, HOEFLE T., FERNANDEZ R., CHARD K., FOSTER I.

Supercomputing

2025

Engine-agnostic Model Hot-swapping for Cost-effective LLM Inference

STOYANOV R., SPIŠAKOVÁ V., REBER A., ARMOUR W., **COPIK M.**, BRUNO R.

CANOPIE-HPC @ Supercomputing

2025

Regular-Confidential LLM Inference: Performance and Cost Across CPU and GPU TEEs

CHRAPEK M., **COPIK M.**, METTAZ E., HOEFLE T.

- Acceptance Rate 41.2% (38/91)

IISWC

2025

DaCe AD: Unifying High-Performance Automatic Differentiation for Machine Learning and Scientific Computing

BOUDAUD A., CALOTOIU A., **COPIK M.**, HOEFLE T.

- Acceptance Rate 20% (39/195)

IEEE Cluster

2025

Cppless: Productive and Performant Serverless Programming in C++

COPIK M., MÖLLER L., CALOTOIU A., HOEFLE T.

ACM TACO

2025

SeBS 2.0: Keeping up with the Clouds

COPIK M., CALOTOIU A., HOEFLE T.

SESAME Workshop @ EuroSys

2025

SeBS-Flow: Benchmarking Serverless Cloud Function Workflows

SCHMID L., **COPIK M.**, CALOTOIU A., BRANDNER L., KOZIOLEK A., HOEFLE T.

- Acceptance Rate in Fall Round 8.2% (30/367)

EuroSys

2025

A Priori Loop Nest Normalization: Automatic Loop Scheduling in Complex Applications

TRÜMPER L., SCHAAD P., ATEB B., CALOTOIU A., **COPIK M.**, HOEFLE T.

CGO

2025

Protocol Buffer Deserialization DPU Offloading in the RPC Datapath

FRANTZ R., GARCIA J., **COPIK M.**, MONROY I., OLMOS J., BLOCH G., DI GIROLAMO S.

IXPUG Workshop @ Supercomputing

2024

Process-as-a-Service: Unifying Elastic and Stateful Clouds with Serverless Processes

COPIK M., CALOTOIU A., GYORGY R., BÖHRINGER R., BRUNO R., HOEFLE T.

- Acceptance Rate 30.1% (63/209)

ACM SoCC

2024

FaaSKeeper: Learning from Building Serverless Services with ZooKeeper as an Example

COPIK M., CALOTOIU A., ZHOU P., TARANOV K., HOEFLE T.

- Acceptance Rate 17.3% (26/150)

ACM HPDC

2024

XaaS: Acceleration as a Service to Enable Productive High-Performance Cloud Computing

HOEFLE T., **COPIK M.**, BECKMAN P., JONES A., FOSTER I., PARASHAR M., REED D., TROYER M., SCHULTHESS T., ERNST D.,

DONGARRA J.

IEEE CISE

2024

Software Resource Disaggregation for HPC with Serverless Computing

COPIK M., CHRAPEK M., SCHMID L., CALOTOIU A., HOEFLE T.

- Acceptance Rate 26.1% (88/337)

IEEE IPDPS

2024

User-guided Page Merging for Memory Deduplication in Serverless Systems

QIU W., **COPIK M.**, WANG Y., CALOTOIU A., HOEFLE T.

- Acceptance Rate 17.5% (92/526)

IEEE Big Data

2023

FMI: Fast and Cheap Message Passing for Serverless Functions

COPIK M., BÖHRINGER R., CALOTOIU A., HOEFLE T.

- Acceptance Rate 29.4% (40/136)

ACM ICS

2023

rFaaS: Enabling High Performance Serverless with RDMA and Leases

COPIK M., TARANOV K., CALOTOIU A., HOEFLE T.

- Acceptance Rate 25.7% (95/369)

IPDPS

2023

Performance-Detective: Automatic Deduction of Cheap and Accurate Performance Models

ACM ICS

SCHMID L., **COPIK M.**, CALOTOIU A., WERLE D., REITER A., SELZER M., KOZIOLEK A., HOEFLE T.

2022

- Acceptance Rate 24.2% (39/161)

MOM: Matrix Operations in MLIR

IMPACT

CHELINI L., BARTHEL H., BIENTINESI P., **COPIK M.**, GROSSER T., SPAMINATO D.

2022

Work-stealing Prefix Scan: Addressing Load Imbalance in Large-scale Image Registration

IEEE TPDS

COPIK M., GROSSER T., HOEFLE T., BIENTINESI P., BERKELS B.

2021

SeBS: A Serverless Benchmark Suite for Function-as-a-Service Computing

ACM/IFIP Middleware

COPIK M., KWASNIEWSKI G., BESTA M., PODSTAWSKI M., HOEFLE T.

2021

- Acceptance Rate 31% (33/107)

Extracting Clean Performance Models from Tainted Programs

ACM PPOPP

COPIK M., CALOTOIU A., GROSSER T., WICKI N., WOLF F., HOEFLE T.

2021

- Acceptance Rate 21% (31/150)

GraphMineSuite: Enabling High-Performance and Programmable Graph Mining Algorithms with Set Algebra

VLDB

BESTA M. [AND 18 OTHERS, INCLUDING **COPIK M.**]

2021

SISA: Set-Centric Instruction Set Architecture for Graph Mining on Processing-in-Memory Systems

IEEE MICRO

BESTA M. [AND 18 OTHERS, INCLUDING **COPIK M.**]

2021

The Generalized Matrix Chain Algorithm

CGO

BARTHEL H., **COPIK M.**, BIENTINESI P.

2018

- Acceptance Rate 28.6% (30/105)

Using SYCL as an Implementation Framework for HPX.Compute

DHPCC++ Workshop, IWOC

COPIK M., KAISER H.

2017

A GPGPU-based Simulator for Prism: Statistical Verification of Results of PMC

CS&P

COPIK M., RATAJ A., WOŻNA-SZCZĘŚNIAK B.

2016

Methods for abdominal respiratory motion tracking

Computer Aided Surgery

SPINCZYK D., KARWAN A., **COPIK M.**

2014

Presentations and Talks

SeBS-Flow: Benchmarking Serverless Cloud Function Workflows

April 2025

EUROSYS 2025 POSTER

High-Performance Serverless: Bridging the Gap Between Elasticity and Efficiency

March 2025

INVITED TALK, VHIVE COMMUNITY MEETING.

XaaS: Acceleration as a Service to Enable Productive High-Performance Cloud Computing

November 2024

PANEL MODERATOR AT CANOPIE-HPC WORKSHOP @ ACM/IEEE SUPERCOMPUTING 2024

Serverless HPC: Challenges, Opportunities, and Future Prospects for Accelerated Cloud Computing

November 2024

PANEL MODERATOR AT ACM/IEEE SUPERCOMPUTING 2024

MIGNificent: Fast, Isolated, and GPU-Enabled Serverless Functions

November 2024

ACM/IEEE SUPERCOMPUTING 2024 POSTER

Benchmarking Serverless with SeBS: Past, Present, and Future

June 2024

THIRD INTERNATIONAL WORKSHOP ON SERVERLESS COMPUTING EXPERIENCE 2024

High Performance Serverless for HPC and Cloud

June 2024

INVITED TALK, INTELLIGENT SERVERLESS AND CLOUD APPLICATIONS SYMPOSIUM, ZURICH UNIVERSITY OF APPLIED SCIENCES.

Evaluating FaaS Systems with the Serverless Benchmark Suite SeBS

June 2024

SEATED WORKSHOP ON SERVERLESS AT THE EDGE, HPDC 2024

Cppless: Productive and Performant Serverless Programming in C++

November 2023

LIGHTNING TALK, LLVM-HPC AT ACM/IEEE SUPERCOMPUTING 2023.

High Performance Serverless for HPC and Clouds

November 2023

POSTER PRESENTATION AT DOCTORAL SHOWCASE, SC 2023.

Serverless As a Bridge Between HPC and Clouds

INVITED TALK, AWS CLOUD FOR RESEARCH AT ETH.

May 2023

Serverless As a Bridge Between HPC and Clouds

INVITED TALK, 5TH WORKSHOP ON PARALLEL AI AND SYSTEMS FOR THE EDGE (PAISE), IPDPS 2023.

May 2023

Serverless As a Bridge Between HPC and Clouds

POSTER PRESENTATION AT PHD FORUM, IPDPS 2023.

May 2023

Software Resource Disaggregation for HPC with Serverless Computing

ACM/IEEE SUPERCOMPUTING 2022 POSTER, **GOLD MEDAL AT THE ACM STUDENT RESEARCH COMPETITION.**

November 2022

Software Resource Disaggregation for HPC with Serverless Computing

SUPERCOMPCLOUD AT ACM/IEEE SUPERCOMPUTING 2022.

November 2022

Interactive Computing with Serverless Functions in rFaaS

LIGHTNING TALK, URGENTHPC AT ACM/IEEE SUPERCOMPUTING 2022.

November 2022

Extracting Clean Performance Models from Tainted Programs”

SIAM CONFERENCE ON PARALLEL PROCESSING FOR SCIENTIFIC COMPUTING (PP22) MINISYMPOSIUM

February 2022

perf-taint: Taint Analysis for Automatic Many-Parameter Performance Modeling

ACM/IEEE SUPERCOMPUTING 2019 POSTER, **GOLD MEDAL AT THE ACM STUDENT RESEARCH COMPETITION.**

November 2019

Parallel Prefix Algorithms for the Registration of Arbitrarily Long Electron Micrograph Series

ACM/IEEE SUPERCOMPUTING 2017 POSTER, ACM STUDENT RESEARCH COMPETITION.

November 2017

HPX and GPU-parallelized STL

C++NOW 2016 CONFERENCE.

May 2016

Service

Program Committee	SoCC 2026, EuroSys 2026, SoCC 2025, CANOPIE-HPC 2025, HiPC 2025, IPDPS 2025, EuroSys 2025 (Shadow PC), SESAME 2025
Organizing Committee	HotCloudPerf 2026, SESAME 2026, PAISE 2025, PAISE 2024
Reviewer	IEEE: TPDS, TC, TCC; Elsevier: Array, Ad-hoc Networks, Journal of Parallel and Distributed Computing, Journal of Systems Architecture, Ocean Modeling; Sage: International Journal of High Performance Computing Applications; LLVM-HPC 2020, ISC 2019
Student Volunteer	ACM/IEEE Supercomputing 2022 & 2023

Teaching

Fall 2024, Fall 2025	Information Systems for Engineers	ETH Zürich
Spring 2025	Information Retrieval	ETH Zürich
Spring 2024	Parallel Programming	ETH Zürich
Fall 2023	Big Data	ETH Zürich
Spring 2023	Parallel Programming	ETH Zürich
Spring 2022	Parallel Programming	ETH Zürich
Fall 2021	Information Systems for Engineers	ETH Zürich
Spring 2021	Parallel Programming	ETH Zürich
Fall 2020	Compiler Design	ETH Zürich
Spring 2020	Parallel Programming	ETH Zürich
Fall 2019	Design of Parallel and High-Performance Computing	ETH Zürich
Spring 2019	Parallel Programming	ETH Zürich
Fall 2018	Numerical Methods for Computational Science and Engineering	ETH Zürich

Students

Ryan Zogg	Bachelor Thesis: Performance Modeling and Serverless Adaptation of HPC Workflows	2025, ETH Zürich
Eiman Alnuaimi	Master Thesis: Acceleration as a Service (XaaS) Source Containers	2025, ETH Zürich

Ritvik Ranjan	Semester Project: Practical Workload Co-location with CRIU and rFaaS	2025, ETH Zürich
Simone Kalbermatter	Co-supervised Semester Project: Conducting a base crypto analysis of the SHArP protocol	2025, ETH Zürich
Andrea Jiang	Co-supervised Master Thesis: Serverless and Cloud Runtimes for Graph-of-Thoughts	2024, ETH Zürich
Entiol Liko	Co-supervised Semester Project: Serverless Co-location with ML	2024, ETH Zürich
Oana Rosca	Semester Project: Long-Term Serverless Performance Variability	2024, ETH Zürich
Constantin Dragancea	Master Thesis: Adoption and Evolution of C++	2024, ETH Zürich
Prajin Khadka	Co-supervised Google Summer of Code Student: Expanded serverless benchmarks	2024, GSoC
Syed Mujtaba	Google Summer of Code Student: Using serverless ZooKeeper in Apache projects	2024, GSoC
Abhishek Kumar	Co-supervised Google Summer of Code Student: New serverless benchmarks	2024, GSoC
Matt Nappo	Co-supervised Google Summer of Code Student: Libfabric Implementation of rFaaS	2023, GSoC
Boyan Zhou	Master Thesis: Adoption and evolution of C++ in HPC Applications	2023, ETH Zürich
Gyorgy Rethy	Master Thesis: Process-as-a-Service computing on modern serverless platforms	2022, ETH Zürich
Laurin Brandner	Master Thesis: Serverless workflows benchmarking	2022, ETH Zürich
Lukas Möller	Bachelor Thesis: Serverless C++ Executor	2022, ETH Zürich
Malte Wächter	Bachelor Thesis: Profiling and optimizations of serverless functions	2022, ETH Zürich
Qiu Wei	Master Thesis: Serverless memory deduplication	2022, ETH Zürich
Lukas Tobler	Master Thesis: Serverless GPU functions	2022, ETH Zürich
Arnet Colin	Bachelor Thesis: Verification of representativeness of benchmarking suite	2021, ETH Zürich
Roman Böhrringer	Master Thesis: Serverless collectives.	2021, ETH Zürich
Emir İşman	Bachelor Thesis: FaaSStest collectives: reliable communication in serverless world	2021, ETH Zürich
Konrad Handrick	Co-supervised Bachelor Thesis: Offloading serverless with sPIN	2021, ETH Zürich
Tobias Lüscher	Bachelor Thesis: TaintImpact: Taint-Based Change Impact Analysis	2021, ETH Zürich
Siegfried Hartogs	Bachelor Thesis: Code-driven Language Development: Framework for Analysis of C/C++ Open-Source Projects	2021, ETH Zürich
Lukas Gygi	Bachelor Thesis: CppBuild: Large-Scale, Automatic Build System for Open Source C++ Repositories	2021, ETH Zürich
Nicolas Wicki	Bachelor Thesis: Control Flow Taint Analysis for Performance Modeling in LLVM	2020, ETH Zürich
Philipp Bomatter	Co-supervised Bachelor Thesis: Towards Extreme-Scale Cache Coherence Protocols and Simulations	2019, ETH Zürich
Gabriel Laberge	Co-supervised Google Summer of Code Student: Alternative smart executors	2018, GSoC
Ajai V George	Google Summer of Code Student: Work on Parallel Algorithms	2017, GSoC